Heat & Thermodynamics

Ideal & Real Gas: Ideal gas is a hypothetical gas which obeys Boyle's law & Charles's law exactly at all temperatures and pressures. But the Real gas does not obey Boyle's law & Charles's law. A real gas is a gas that does not behave as an ideal gas due to interactions between gas molecules. Some differences between Ideal gas and Real gas-

- Ideal gas has no definite volume while real gas has definite volume.
- Ideal gas has no mass whereas real gas has mass.
- Collision of ideal gas particles is elastic while non-elastic for real gas.
- No energy involved during collision of particles in ideal gas. Collision of particles in real gas has attracting energy.
- Pressure is high in ideal gas compared to real gas.
- Ideal gas follows the equation- $\boldsymbol{PV = nRT}$, Real gas follows the equation-
    - $\boldsymbol{(P + \frac{n^2a}{V^2})(V - nb) = nRT}$ (Van-der Waals equation)

Van der Waal's Equation of State: At low temperatures or high external pressures, real gases deviate significantly from ideal gas behavior. *Van der Waals* realized that two of the assumptions of the kinetic molecular theory were questionable. The kinetic theory assumes that-

- Gas particles occupy a negligible fraction of the total volume of the gas.
- The force of attraction between gas molecules is zero.

At low temperatures or high external pressures, these are no longer true. Some Corrections must be needed for the Real Gases.

***Volume Correction**:* The molecules of an ideal gas are assumed to have zero volume; the volume available to them for motion is always the same as the volume of the container. But At low temperatures and high external pressures, the intermolecular distances become smaller and smaller. As a result, the volume occupied by the molecules becomes significant compared with the volume of the container. Consequently, the volume available for motion is not equal to the volume of the container. The volume of the Real gas will be the volume of the container minus the summation of volume of all molecules itself.

$$PV = nRT \qquad (for\ Ideal\ gas)$$

$$P(V - nb) = nRT \qquad (for\ Real\ gas\ after\ volume\ correction)$$

***Pressure Correction**:* The assumption that there is no force of attraction between gas molecules or between the molecules and the walls of the container. If it was, gases would never condense to form liquids. In reality, there is a small force of attraction between gas molecules together and also with the walls of the container.At low temperatures and high external pressures, the intermolecular distances become smaller and smaller, the kinetic energy of the gas molecules decreases. Eventually, a point is reached where the molecules can no longer overcome the intermolecular attractive forces, and the gas liquefies (condenses to a liquid).These attractions lessen the force with which the molecule hits the wall. As a result, the pressure is less than that of an ideal gas.

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$$PV = nRT \qquad (for\ Ideal\ gas)$$

$$P = \frac{nRT}{(V-nb)} - \frac{n^2a}{V^2} \qquad (for\ Real\ gas\ afterPressure\ Correction)$$

$$\boldsymbol{\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT(for\ Real\ gas)}$$

First Law of Thermodynamics: According to famous scientist ***Joule***-

"When work is transformed into heat or heat is transformed into work, then work and heat is directly proportional to each other."

If Q be the amount of heat produced due to transformation of W amount of work, then according to the first law of thermodynamics-

$$W \propto Q$$

$$W = J\,Q$$

Here, J is proportionality constant. It is called mechanical equivalent of heat or Joules equivalent.

According to ***Clausius***-

"*If some heat supplied to a system which can do work, then the quantity of heat absorbed by the system is equal to the sum of the increase in the internal energy of the system and the external work done by the system"*

If dQ be the amount of heat absorbed by the system and dU be the change in the internal energy and dW, the work done by the system due to the absorption of heat, then according to the first law of thermodynamics-

$$dQ = dU + dW$$

$$\boldsymbol{dQ = dU + Pdv}$$

Some Definitions:

- **Internal Energy (*dU*)**: the sum of the kinetic energy and potential energies of the molecules of a system is called its Internal Energy (dU). It depends only on the temperature but not on pressure or volume.
- **Specific Heat**: The amount of heat needed to increase the temperature of a substance of unit mass by 1K is called Specific heat of that substance.

$$s = \frac{dQ}{mdT}$$

  - At constant volume, molar specific heat- $C_v = \frac{dQ}{dT}$ (for 1 mole)
  - At constant Pressure, molar specific heat- $C_p = \frac{dQ}{dT}$ (for 1 mole)

- Different thermo-dynamical changes:
  - Isothermal change- $dT = 0;\quad T = Constant$
  - Adiabatic change- $dQ = 0;\quad Q = Constant$
  - Isochronic change- $dV = 0;\quad V = Constant$
  - Isobaric change- $dP = 0;\quad P = Constant$

<u>Distinction between Isothermal and Adiabatic changes</u>:

Isothermal change	Adiabatic change
1. Change of pressure and volume of a gas at constant temperature is called isothermal change.	1. Change of pressure and volume of a gas by keeping the total heat content of the gas fixed is called adiabatic change.
2. In this change necessary heat is taken or rejected to keep the temperature constant.	2. Change of temperature takes place in this process.
3. It is a very slow process.	3. It is a very fast process.
4. The container of the gas should be good conductor of heat.	4. The container of the gas should be non-conducting.
5. The heat capacity of the surrounding is very high.	5. The heat capacity of the surrounding should be low.
6. It follows Boyle's Law	6. It does not follow Boyle's Law.
7. Isothermal curve is less steeper then adiabatic curve.	7. Adiabatic curve is more steeper than isothermal curve.

<u>Relation between C_v & C_p:</u>

Let us consider a cylinder C with a frictionless movable piston D [Fig. 1]. Both the cylinder and the piston are bad conductors of heat. Consider that 1 moles of ideal gas is kept in the cylinder at volume V and pressure P. The temperature of the gas is raised by dT, by supplying heat dQ.

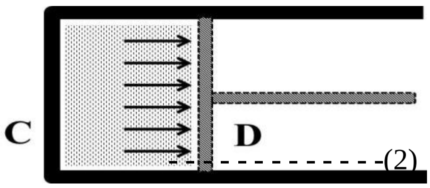
According to the first law of thermodynamics-

$$dQ = dU + dW = dU + PdV \quad \text{-------------(1)}$$

At constant volume, $dV = 0; dU = dQ = C_v dT$ (for 1 mole)

At constant Pressure, we can be written as- $dQ = C_p dT$ (for 1 mole)

Fig. 1



Inserting the values of dU and dQ in equation (1) we get,

$$C_p dT = C_v dT + PdV \quad \text{-------------(2)}$$

The ideal gas equation for 1 mole- $PV = RT$; Differentiating this equation at constant pressure we get,

$$PdV = RdT$$

Then, equation (2),

$$C_p dT = C_v dT + RdT$$

$$C_p dT - C_v dT = RdT \quad \text{-------------(2)}$$

$$\boldsymbol{C_p - C_v = R}$$

Informations: ***i)*** $\boldsymbol{C_p > C_v}$ ***ii)*** $\boldsymbol{\gamma = \frac{C_p}{C_v}}$

For monatomic gas $\boldsymbol{\gamma = 1.66}$; ***For diatomic gas*** $\boldsymbol{\gamma = 1.40}$; ***For triatomic gas*** $\boldsymbol{\gamma = 1.33}$

<u>Relation between P, V & T of a gas in Adiabatic Change</u>: Let dQ amount of heat supplied in 1mole of an ideal gas. Due to this temperature of the gas will increase and the gas will do some work i.e., the supplied heat will be used in two ways. Let the change in volume and temperature be dV and dT respectively.

So, from the first law of thermodynamics we get,

$$dQ = C_v dT + PdV \qquad \text{-------------(1)}$$

Here, $C_v = specific\ heat\ of\ gas\ at\ constant\ volume,$ and $PdV =$ work done due to the expansion of the gas at constant pressure.

We know, in adiabatic process no heat exchange takes place with the surroundings, so-

$$dQ = 0$$

From Eq. 1. We get, $\quad C_v dT + PdV = 0 \qquad$ -------------(2)

Again for an ideal gas, $PV = RT,$ here R is molar gas constant.

Differentiating the above relation we get,

$$PdV + VdP = RdT$$

$$or, \quad dT = \frac{PdV + VdP}{R}$$

From Eq.2.we get,

$$C_v\left(\frac{PdV + VdP}{R}\right) + PdV = 0$$

or, $\quad C_v PdV + C_v VdP + RPdV = 0$

or, $\quad C_v PdV + C_v VdP + (C_P - C_v)PdV = 0 \qquad [\because C_P - C_V = R]$

or, $\quad C_v PdV + C_v VdP + C_P PdV - C_v PdV = 0$

or, $\quad C_v VdP + C_P PdV = 0$

or, $\quad VdP + \frac{C_P}{C_v}PdV = 0 \qquad$ [Dividing by C_V]

or, $\quad VdP + \gamma PdV = 0 \qquad [\because \frac{C_P}{C_V} = \gamma]$

or, $\quad \frac{dP}{P} + \gamma\frac{dV}{V} = 0 \qquad$ [Dividing by PV]

Now integrating above equation, we get,

$$lnP + \gamma lnV = Constant$$

or, $\quad ln(PV^{\gamma}) = Constant$

$\therefore \quad \boldsymbol{PV^{\gamma} = Constant} \quad$ So that, $\boldsymbol{P_1V_1^{\gamma} = P_2V_2^{\gamma}}$

Using ideal gas equation, $PV = RT$; We can be also written as-

- $\boldsymbol{TV^{\gamma-1} = Constant;} \qquad \boldsymbol{T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}}$

- $\boldsymbol{T^{\gamma}P^{1-\gamma} = Constant;} \qquad \boldsymbol{T_1^{\gamma}P_1^{1-\gamma} = T_2^{\gamma}P_2^{1-\gamma}}$

<u>Characteristics of the Adiabatic and Isothermal Curve</u>: Both isothermal and adiabatic process can be represented by pressure versus volume graphs as shown in Fig. 1. The tangent drawn at any point in the graph gives the slope $\frac{dP}{dV}$ · it is seen that at any point the slope of adiabatic curve is γ times greater than the slope of the isothermal curve.

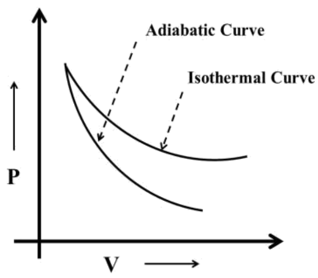


Fig. 1

We know, the ideal gas equation for isothermal case,

$$PV = Constant \qquad \text{-----(1)}$$

Differentiating equation (1) we get, $\quad PdV + VdP = 0$

$$\text{or,} \quad \left(\frac{dP}{dV}\right)_{isothermal} = -\frac{P}{V} \qquad \text{-------------(2)}$$

Again, in adiabatic case, the ideal gas equation,

$$PV^{\gamma} = Constant \qquad \text{-------------(3)}$$

Differentiating equation (3) we get, $\quad \gamma PV^{\gamma}dV + V^{\gamma}dP = 0$

$$\text{or,} \quad \left(\frac{dP}{dV}\right)_{adiabatic} = -\frac{\gamma PV^{\gamma-1}}{V^{\gamma}} = -\gamma\frac{P}{V} \qquad \text{-------------(4)}$$

Comparing equations (2) and (4) we find,

$$\boldsymbol{\left(\frac{dP}{dV}\right)_{adiabatic} = \gamma\left(\frac{dP}{dV}\right)_{isothermal}}$$

So, $\boldsymbol{Slope\ of\ the\ adiabatic = \gamma \times Slope\ of\ the\ isothermals}$

Since, $\gamma > 1$; hence, at any point the slope of adiabatic curve is γ times greater than the slope of the isothermal curve.

Problem-1: *For an ideal gas* $\gamma = 1.4$,*calculate the values of molar specific heats of the gas.* *(*$R = 8.31\, Jmol^{-1}K^{-1}$)

Problem-2: *When* $5 \times 10^3 J$ *heat is supplied to a gas of volume* $100m^3$ *at normal pressure, the volume changes to*$100.2\, m^3$. *Find the work done by the gas and change in internal energy.*

Problem-3: *The volume of air is made three times the original volume by adiabatic expansion. If the initial pressure is 1 atmospheric pressure, what is the final pressure? [*$\gamma = 1.40$*]*

Problem-4: *A fixed amount of gas at 27*^{0}*C is suddenly expanded to double its original volume. What is the final Temperature? [Name of the gas is **Helium**]*